resolve_cross_validation_fold_search.R

October 6, 2026

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`resolve_cross_validation_fold_search` *Resolve a Feasible Grouped Cross-Validation Design*

Description

Preserves feasible initial assignments, otherwise tests increasing grouped fold counts before the existing leave-one-location-out fallback. No models are fitted during this search and no feasibility threshold is relaxed.

Usage

```
resolve_cross_validation_fold_search(
  data_locations = NULL,
  data_assignments = NULL,
  data_partition_diagnostics = NULL,
  data_community_matrix = NULL,
  config_data_processing = NULL,
  config_cross_validation = NULL,
  grouped_fold_ladder = base::c(5L, 10L, 20L, 50L),
  verbose = TRUE
)
```

Arguments

`data_locations`, `data_assignments`, `data_partition_diagnostics`
: Initial location, assignment, and training-partition diagnostic tables.

`data_community_matrix`
: Binary response matrix aligned with location row indices.

`config_data_processing`
: Processing configuration with ‘min_n_cores‘, ‘min_n_samples‘, ‘min_n_taxa‘.

`config_cross_validation`
: CV controls including MEM threshold, repeats, assignment seed, and grid.

`grouped_fold_ladder`
: Strictly increasing unique integer fold counts, all at least two.

`verbose`
: Logical. Print selected design and candidate-fold task estimates if TRUE.

Value

Named list containing unchanged-schema 'data_assignments', attempt-level 'data_fold_search', partition-level 'data_partition_diagnostics_search', and the configured 'vec_grouped_fold_ladder'. Failed grids are recorded; unexpected errors propagate. All configured repeats must pass. The search never tries counts at or above location count.

Examples

```
## Not run:
list_search <-
  resolve_cross_validation_fold_search(
    data_locations = data_cross_validation_locations,
    data_assignments = data_cross_validation_assignments_initial,
    data_partition_diagnostics =
      data_cross_validation_partition_diagnostics_initial,
    data_community_matrix = data_community_filtered,
    config_data_processing = config_data_processing,
    config_cross_validation = config_model_fitting[["cross_validation"]]
  )

## End(Not run)
```

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